

DOES URBANIZATION FILTER BIRDS ON THE BASIS OF THEIR BIOLOGICAL TRAITS?

SOLÈNE CROCI,^{1,2,4} ALAIN BUTET,² AND PHILIPPE CLERGEAU^{1,3}

¹INRA SCRIBE, Campus de Beaulieu, Av. Général Leclerc, 35042 Rennes, France

²CAREN UMR CNRS 6553 ECOBIO Univ. Rennes 1, Campus de Beaulieu, Av. Général Leclerc, 35042 Rennes, France

³Muséum National d'Histoire Naturelle, Département Ecologie et Gestion de la Biodiversité-UMR 5173, 55 rue Buffon, 75005 Paris, France

Abstract. Urbanization leads to the biotic homogenization of global avifauna. We hypothesized that urbanization acts as a filter on species traits and, therefore, that urban passerines share biological traits explaining their capacity to tolerate urban constraints. We investigated 18 biological traits of passerines related to their general biology, distribution, breeding, and morphometry. In a regional analysis conducted on passerine data from one Swiss and 11 French cities (regional analysis), we identified urban adapters (tolerant species) and urban avoiders (intolerant species), and compared their traits. In a local analysis conducted on passerine data of 13 woodlands located along a short rural-urban gradient, we identified groups of species associated with particular vegetation structures within or particular landscape structures around woodlands. We associated each of these species groups with a tolerance level to urbanization and compared their traits. Regional analysis revealed that urban adapters prefer forest environments, are sedentary, omnivorous, widely distributed, high-nesters with large wingspans. Urban avoiders seem to allocate more energy to reproduction than do urban adapters, to the detriment of adaptation to new environments such as urban areas. Local analysis did not reveal any link between traits and species tolerance levels. At large spatial scales, urbanization seems therefore to act as a filter on species traits. However, the urban constraints that filter species at such large scales do not seem to be the same ones that determine species distribution at local scales. Analyses of traits are powerful tools to understanding regional community composition between urban and rural areas.

Key words: biological filter, biological traits, environmental tolerance, functional homogenization, passerines, urbanization, woodlands.

¿Actúa la Urbanización como un Filtro para las Aves de acuerdo a sus Rasgos Biológicos?

Resumen. La urbanización conduce a la homogenización biótica de la avifauna global. Nosotros hipotetizamos que la urbanización actúa como un filtro sobre los rasgos de las especies, y, por lo tanto, que las aves paserinas urbanas comparten rasgos biológicos que explican su capacidad para tolerar las limitaciones impuestas por el ambiente urbanizado. Investigamos 18 rasgos de las aves paserinas relacionados con su biología general, distribución, reproducción y morfometría. En un análisis regional realizado con datos de aves paserinas de una ciudad suiza y 11 ciudades francesas, identificamos especies que se adaptan al ambiente urbano (especies tolerantes) y especies que lo evitan (especies intolerantes), y comparamos sus rasgos. En un análisis local realizado con datos de aves paserinas de 13 bosques ubicados a lo largo de un gradiente rural-urbano corto, identificamos grupos de especies asociadas con estructuras de vegetación particulares del bosque o con características del paisaje circundante a los bosques. Asociamos cada uno de esos grupos de especies con un nivel de tolerancia a la urbanización y comparamos sus rasgos. El análisis regional reveló que las especies que se adaptan al ambiente urbano prefieren ambientes boscosos, son sedentarias, son omnívoras, presentan amplia distribución, anidan en lugares altos y presentan envergaduras grandes. Las especies que evitan las ciudades parecen asignar más energía a la reproducción que las especies tolerantes, lo que va en detrimento de su adaptación a nuevos ambientes como las áreas urbanas. El análisis local no mostró ninguna conexión entre los rasgos de las especies y su nivel de tolerancia. Por lo tanto, a escalas espaciales grandes, la urbanización parece actuar como un filtro sobre los rasgos de las especies. Sin embargo, las restricciones del ambiente urbano que filtran las especies a esas escalas grandes no parecen ser las mismas que determinan la distribución de las especies a escalas locales. Los análisis de rasgos biológicos representan una herramienta útil para entender la composición regional de las comunidades entre áreas urbanas y rurales.

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⁴Present address: UMR 5023 Ecology of Fluvial Hydrosystems (Bat. Darwin C) University Lyon 1, 43 Boulevard du 11 Novembre 1918, 69622 Villeurbanne cedex, France. E-mail: solene.croci@gmail.com

INTRODUCTION

Urbanization leads to very similar forms throughout the world (Savard et al. 2000, Clergeau et al. 2001, McKinney 2006) such as roads, buildings, and other structures needed by human populations, which are developed to the detriment of the natural environment. This process has diverse ecological effects such as fragmentation, isolation, and loss of natural habitat patches, as well as creation of new habitats (Marzluff and Ewing 2001, Melles et al. 2003, Loram et al. 2007). In addition, urbanization induces disturbances due to human activity (Fernández-Juricic and Schroeder 2003), heat islands (Voogt and Oke 2003), pollution (Fenger 1999), and noise (Slabbekoorn et al. 2007). Therefore, urban areas seem a priori to offer few favorable conditions for wildlife. However, some species thrive in such altered environments.

Studies on urbanization's impact on biodiversity along rural-urban gradients, and especially on bird communities, have shown a decrease in species richness with increasing urbanization (Blair 1996, Clergeau et al. 1998, Clergeau, Croci, et al. 2006, McKinney 2006) and changes in species composition from natural areas to more urbanized landscapes. Some bird species, referred to as urban adapters, seem tolerant to urban constraints and are able to maintain populations in urban areas, whereas many other urban avoiders appear intolerant (Blair 1996). The identification of these two groups of species has led ecologists to search for what characters make a bird an urban adapter or an urban avoider. They have shown that these two kinds of species differ in biological traits such as feeding habitat, diet, nesting, migratory behavior, and behavioral flexibility (McClure 1989, Lim and Sodhi 2004, Clergeau, Croci, et al. 2006, Kark et al. 2007), and thus that urbanization leads to functional homogenization of birds communities (Blair 2001, Crooks et al. 2004, Olden et al. 2004, Clergeau, Croci, et al. 2006, McKinney 2006).

Following Clergeau, Croci, et al. (2006), we postulated that urban constraints seem to act as a filter that selects species on the basis of their biological traits. We considered that species tolerance to new environments partly results from a combination of several of these traits. We therefore expected that urban adapters should have several specific traits that might favor them in urban centers. Our aim was therefore to identify which traits characterize urban adapters and urban avoiders and then, to analyze in which ways those traits could provide advantage or disadvantage to species in urban areas.

We conducted explanatory research of traits that could enable species to tolerate urbanized areas by selecting both biological traits already highlighted in previous studies and new, original traits potentially linked to passerine tolerance to new environments. In parallel, we identified different species tolerance patterns to urban constraints using results from two scale analyses. First, at a regional scale, we compared biological traits between species tolerant or intolerant to urbanization. These opposing tolerance levels were assessed by listing bird occurrences in the centers and surrounding areas of one Swiss and 11 French cities. Second, because tolerance patterns

to urban constraints can be sufficiently complex to necessitate description beyond simply tolerant or intolerant, we compared, at a local scale, the biological traits of species distributed in woodlands stretching from the center to close surroundings (i.e., within 10 km) of one French city. We considered that each distribution pattern corresponded to a particular tolerance level. We focused on a single habitat type—woodland—because habitat availability in a city and within close proximity can act as a covariate with observed bird distributions along a rural-urban gradient (Jokimäki and Kaisanlahti-Jokimäki 2003, Clergeau, Jokimäki, and Snep 2006, Sadler et al. 2006).

We hypothesized that our analyses would allow us to identify several combined traits that discriminate bird species according to their tolerance to urbanization. Consequently, the aims of our study were to analyze how some combined biological traits (1) could help to predict which species will tolerate urban areas as opposed to those that will not (regional analysis), and (2) could facilitate the understanding of the species distribution from a town center to its rural surroundings (local analysis).

METHODS

REGIONAL ANALYSIS: URBAN ADAPTERS VERSUS URBAN AVOIDERS

In order to test our hypothesis at a regional scale, we compared birds able to tolerate urban constraints (i.e., to breed at town center), the urban adapters, to those that appear unable to do so, the urban avoiders. A list of urban adapters was compiled from the data from ten publications based in 12 cities of different sizes and representative of medium-sized towns in France and Switzerland (Fig. 1): Dijon (Ferry and Ferry 1965), Lausanne (Chessex and Ribaut 1966, Tschanz et al. 1993), Caen (Collette 1971, Noël 1998), Marseille (Marchetti 1976), Lyon (Tatibouët 1981), Nîmes (Gory 1992), Angers, Rennes, Nantes (Clergeau et al. 2001) and Laval, Versailles, Melun (Croc 2003; for details see Croci et al. 2007). In order to take into account species exhibiting real tolerance to urban constraints, we did not consider data from sites located at the edges of the cities in these studies. All passerines reported to breed in urban centers in previous studies were classified as urban adapters (Table 1).

A list of regional species was compiled from the atlas of Yeatman-Berthelot and Jarry (1994). This atlas provides maps of France divided into 1092 units where the breeding of species is recorded as certain, possible, probable, or not recorded. We defined an area of four units (i.e., 2592 km²) around each of the 12 focal cities. All species whose breeding was certain or possible in one of the four units were included in the regional species pool. Data for Lausanne were available in this grid as well. We then compared this list to our list of urban adapters. All species breeding in urban centers were also found in the areas surrounding the cities. All passerines present in the regional pool and not recorded as urban adapters were classified as urban avoiders (Table 1).

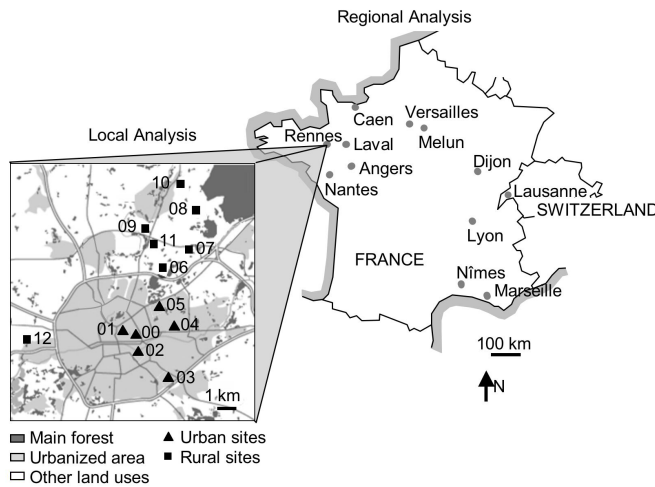


FIGURE 1. Location of the surveyed towns and sites. The regional analysis was conducted on avifauna data compiled from 12 towns and their surroundings (from north to south): Caen, Versailles, Melun, Rennes, Laval, Angers, Dijon, Nantes, Lausanne, Lyon, Nîmes, and Marseille. The local analysis was conducted in 2004 and 2005 on avifauna surveyed in 13 woodlands distributed along a rural-urban gradient extending 10 km from the center of Rennes to its more rural adjacent landscape. Six sites are located in town, and seven are located in its surroundings.

LOCAL ANALYSIS: INDICATOR SPECIES ALONG A RURAL-URBAN GRADIENT

We conducted a narrow spatial scale analysis of 13 small woodlands (0.5 to 2 ha) distributed along a gradient of urbanization (Fig. 1). This gradient extended over 10 km from the city center of Rennes to its more rural adjacent landscapes. Six sites (sites 0–5) were in the city and corresponded to wooded areas in parks (sites 0, 1, and 4), a wooded lane (site 2), or small urban woodlands (sites 3 and 5). Seven sites were in the adjacent rural landscape (sites 6–12) and corresponded to small woodlands included in more- or less-open landscapes, ranging from a hedgerow network (site 8) to open fields (site 10). Site locations were chosen because they were the most representative of landscape diversity found within and outside French cities.

In each woodland, we conducted one point count (Bibby et al. 2000) per site every three weeks from April to July in 2004 and 2005, for a total of ten point counts per site. Each point count lasted 20 min, during which every bird heard or seen was determined at the species level and counted. Surveys were always conducted in the early morning during the week (between 07:30 and 10:00 [UTC+2]) by the same person. We were careful to avoid double counts of the same individual birds and limited the survey to woodland areas. Bird densities measured at each point count were averaged across all point counts ($n = 10$) per site.

A habitat analysis was conducted within each woodland. We used different measures to describe the vegetation structure: the maximum species richness (herbaceous richness)

and the Shannon diversity index (herbaceous diversity) of herbaceous flora, the percentage of bare ground cover (bare ground), the percentage of litter cover (litter cover), the percentage of herbaceous cover (vegetation of 0–1 m height; herbaceous cover), the percentage of shrubby cover (>1–4 m; shrubby cover), and the percentage of tree cover (>4 m; tree cover). In parallel, for a landscape analysis using GIS (Arcview 8.3, ESRI, Redlands, California) and Fragstats Version 3.0 (McGarigal et al. 2002), we extracted variables describing landscape composition and configuration in a buffer zone with a radius of 600 m around each woodland: percent of soil occupied by four land use classes including crops (crop), wood or forest (wood), lawn or grass (grass), mineral surfaces such as roads and buildings (mineral surface); the Shannon diversity index of buffer zones (land use diversity) as a measure of land use diversity; mean edge density (edge density) and mean patch area (patch area) as measures of fragmentation; mean Euclidean nearest neighbor distance (patch proximity) to determine distances between patches of the same land use classes; mean shape index (patch shape) to describe patch thickness or regularity; and mean wood cohesion index (wood connectivity) as a measure of connectivity between patches of wood. Using ADE4 (Thioulouse et al. 1997), principal component analysis (PCA), and hierarchical ascendant classification (HAC) were conducted on each of these two sets of variables to identify groups of similar sites in terms of their habitat or surrounding landscape structure.

Then, we conducted an IndVal analysis (Dufrêne and Legendre 1997) on the mean densities of bird species at each site and on site groups highlighted in previous analyses. This analysis allowed identification of indicator species of group of sites. We considered species identified as indicators of the same group of sites (i.e., having the same unequal distribution pattern along the gradient) to have the same capacity to tolerate urbanization. Such species were associated with the same tolerance status (Table 1), which depends on the habitat within (H) or landscape around (L) the sites they indicated; this tolerance status can be especially high (H- or L-very urbanophilic) or low (H- or L-moderately urbanophilic). We considered species not identified as indicators of a group of sites (i.e., species equally distributed along the gradient) to have a medium capacity to tolerate urbanization and to be associated with a medium tolerance status (H- or L-urbanophilic).

BIRD BIOLOGICAL TRAITS

We compiled information on bird traits from different atlases and guides (Géroutet 1957–1972, 1961, 1963, Voous 1960, Yeatman-Berthelot and Jarry 1994, Mullarney et al. 1999, Dubois et al. 2001, Hume et al. 2003; Table 2). We focused on traits either linked or potentially linked to species tolerance to new environments such as urban areas. Traits were either qualitative (i.e., given as descriptions in atlases; e.g., omnivorous diet), or quantitative (i.e., given as numerical values; e.g.,

TABLE 1. List of species surveyed at the regional ($n = 110$) and the local ($n = 31$) scales, following phylogenetic order of Dubois et al. (2001). Species are associated with different tolerance statuses: urban avoiders or urban adapters for the regional analysis; very urbanophilic, moderately urbanophilic or urbanophilic for the local analysis, according to description of habitat within woodlands (H-) or landscape around woodlands (L-). “Regional group membership” and “Local group membership” columns indicate in which class the species belong according to regional and local cluster analyses, respectively.

Family Species	Regional analysis		Local analysis		
	Regional urban tolerance	Regional group membership	Landscape tolerance status	Habitat tolerance status	Local group membership
Alaudidae					
<i>Melanocorypha calandra</i> (Calandra Lark)	Avoider	1			
<i>Calandrella brachydactyla</i> (Short-toed Lark)	Avoider	1			
<i>Galerida cristata</i> (Crested Lark)	Avoider	1			
<i>Lullula arborea</i> (Woodlark)	Avoider	1			
<i>Alauda arvensis</i> (Skylark)	Avoider	1			
Hirundinidae					
<i>Riparia riparia</i> (Sand Martin)	Avoider	2			
<i>Ptyonoprogne rupestris</i> (Crag Martin)	Avoider	2			
<i>Hirundo rustica</i> (Barn Swallow)	Adapter	2	L-urbanophilic	H-very urbanophilic	2
<i>Delichon urbica</i> (House Martin)	Adapter	2			
Motacillidae					
<i>Anthus campestris</i> (Tawny Pipit)	Avoider	1			
<i>Anthus trivialis</i> (Tree Pipit)	Avoider	1			
<i>Anthus pratensis</i> (Meadow Pipit)	Avoider	1			
<i>Anthus spinoletta</i> (Rock Pipit)	Avoider	1			
<i>Motacilla flava</i> (Yellow Wagtail)	Avoider	1			
<i>Motacilla cinerea</i> (Grey Wagtail)	Adapter	1			
<i>Motacilla alba</i> (Pied Wagtail)	Adapter	1			
Cinclidae					
<i>Cinclus cinclus</i> (Dipper)	Avoider	1			
Troglodytidae					
<i>Troglodytes troglodytes</i> (Winter Wren)	Adapter	2	L-urbanophilic	H-urbanophilic	2
Prunellidae					
<i>Prunella modularis</i> (Dunnock)	Adapter	1	L-urbanophilic	H-urbanophilic	1
<i>Prunella collaris</i> (Alpine Accentor)	Avoider	1			
Turdidae					
<i>Erithacus rubecula</i> (European Robin)	Adapter	1	L-urbanophilic	H-moderately urbanophilic	1
<i>Luscinia megarhynchos</i> (Nightingale)	Adapter	1			
<i>Luscinia svecica</i> (Bluethroat)	Avoider	1			
<i>Phoenicurus ochruros</i> (Black Redstart)	Adapter	2			
<i>Phoenicurus phoenicurus</i> (Redstart)	Adapter	1			
<i>Saxicola rubetra</i> (Whinchat)	Avoider	1			
<i>Saxicola torquata</i> (Stonechat)	Avoider	1			
<i>Oenanthe oenanthe</i> (Wheatear)	Avoider	1			
<i>Oenanthe hispanica</i> (Black-eared Wheatear)	Avoider	1			
<i>Monticola saxatilis</i> (Rock Thrush)	Avoider	1			
<i>Monticola solitarius</i> (Blue Rock Thrush)	Adapter	2			
<i>Turdus torquatus</i> (Ring Ouzel)	Avoider	2			
<i>Turdus merula</i> (Blackbird)	Adapter	1	L-very urbanophilic	H-urbanophilic	1
<i>Turdus pilaris</i> (Fieldfare)	Avoider	2			
<i>Turdus philomelos</i> (Song Thrush)	Adapter	1	L-very urbanophilic	H-urbanophilic	1
<i>Turdus viscivorus</i> (Mistle Thrush)	Adapter	1	L-urbanophilic	H-urbanophilic	1
Sylviidae					
<i>Cettia cetti</i> (Cetti's Warbler)	Avoider	1			
<i>Cisticola juncidis</i> (Fan-tailed Warbler)	Avoider	1			
<i>Locustella naevia</i> (Grasshopper Warbler)	Avoider	1			
<i>Locustella luscinioides</i> (Savi's Warbler)	Avoider	1			
<i>Acrocephalus schoenobaenus</i> (Sedge Warbler)	Avoider	1			
<i>Acrocephalus palustris</i> (Marsh Warbler)	Avoider	1			

(continued)

TABLE 1. (Continued)

Family Species	Regional analysis		Local analysis		
	Regional urban tolerance	Regional group membership	Landscape tolerance status	Habitat tolerance status	Local group membership
<i>Acrocephalus scirpaceus</i> (Reed Warbler)	Avoider	1			
<i>Acrocephalus arundinaceus</i> (Great Reed Warbler)	Avoider	1			
<i>Hippolais icterina</i> (Icterine Warbler)	Avoider	1			
<i>Hippolais polyglotta</i> (Melodious Warbler)	Avoider	1			
<i>Sylvia undata</i> (Dartford Warbler)	Avoider	1			
<i>Sylvia conspicillata</i> (Spectacled Warbler)	Avoider	1			
<i>Sylvia cantillans</i> (Subalpine Warbler)	Avoider	1			
<i>Sylvia melanocephala</i> (Sardinian Warbler)	Adapter	1			
<i>Sylvia hortensis</i> (Orphean Warbler)	Avoider	1			
<i>Sylvia curruca</i> (Lesser Whitethroat)	Avoider	1			
<i>Sylvia communis</i> (Whitethroat)	Avoider	1			
<i>Sylvia borin</i> (Garden Warbler)	Avoider	1			
<i>Sylvia atricapilla</i> (Blackcap)	Adapter	1	L-urbanophilic	H-urbanophilic	1
<i>Phylloscopus bonelli</i> (Bonelli's Warbler)	Avoider	1			
<i>Phylloscopus sibilatrix</i> (Wood Warbler)	Adapter	1			
<i>Phylloscopus collybita</i> (Chiffchaff)	Adapter	1	L-moderately urbanophilic	H-moderately urbanophilic	1
<i>Phylloscopus trochilus</i> (Willow Warbler)	Adapter	1	L-urbanophilic	H-urbanophilic	1
<i>Regulus regulus</i> (Goldcrest)	Adapter	2	L-very urbanophilic	H-urbanophilic	2
<i>Regulus ignicapillus</i> (Firecrest)	Adapter	2	L-urbanophilic	H-very urbanophilic	2
Muscicapidae					
<i>Muscicapa striata</i> (Spotted Flycatcher)	Adapter	1			
<i>Ficedula hypoleuca</i> (Pied Flycatcher)	Avoider	1			
Timaliidae					
<i>Panurus biarmicus</i> (Bearded Tit)	Avoider	1			
Aegithalidae					
<i>Aegithalos caudatus</i> (Long-tailed Tit)	Adapter	2	L-urbanophilic	H-urbanophilic	2
Paridae					
<i>Parus palustris</i> (Marsh Tit)	Adapter	2	L-urbanophilic	H-urbanophilic	2
<i>Parus montanus</i> (Willow Tit)	Adapter	2			
<i>Parus cristatus</i> (Crested Tit)	Adapter	2	L-urbanophilic	H-urbanophilic	2
<i>Parus ater</i> (Coal Tit)	Adapter	2	L-urbanophilic	H-urbanophilic	2
<i>Parus caeruleus</i> (Blue Tit)	Adapter	2	L-moderately urbanophilic	H-moderately urbanophilic	2
<i>Parus major</i> (Great Tit)	Adapter	2	L-moderately urbanophilic	H-urbanophilic	2
Sittidae					
<i>Sitta europaea</i> (Nuthatch)	Adapter	2	L-urbanophilic	H-urbanophilic	2
Tichodromadidae					
<i>Tichodroma muraria</i> (Wallcreeper)	Avoider	2			
Certhiidae					
<i>Certhia familiaris</i> (Treecreeper)	Avoider	2			
<i>Certhia brachydactyla</i> (Short-toed Treecreeper)	Adapter	2	L-urbanophilic	H-urbanophilic	2
Oriolidae					
<i>Oriolus oriolus</i> (Golden Oriole)	Avoider	1			
Laniidae					
<i>Lanius collurio</i> (Red-backed Shrike)	Avoider	1			
<i>Lanius minor</i> (Lesser Grey Shrike)	Avoider	1			
<i>Lanius excubitor</i> (Great Grey Shrike)	Avoider	2			
<i>Lanius senator</i> (Woodchat Shrike)	Avoider	1			
Corvidae					
<i>Garrulus glandarius</i> (Jay)	Adapter	2	L-urbanophilic	H-urbanophilic	2

(continued)

TABLE 1. (Continued)

Family Species	Regional analysis		Local analysis		
	Regional urban tolerance	Regional group membership	Landscape tolerance status	Habitat tolerance status	Local group membership
<i>Pica pica</i> (Magpie)	Adapter	2	L-very urbanophilic	H-urbanophilic	2
<i>Nucifraga caryocatactes</i> (Nutcracker)	Avoider	2			
<i>Pyrrhocorax graculus</i> (Alpine Chough)	Avoider	2			
<i>Corvus monedula</i> (Jackdaw)	Adapter	2			
<i>Corvus frugilegus</i> (Rook)	Adapter	2			
<i>Corvus corone</i> (Carrion Crow)	Adapter	2	L-urbanophilic	H-urbanophilic	2
<i>Corvus corax</i> (Raven)	Avoider	2			
Sturnidae					
<i>Sturnus vulgaris</i> (Starling)	Adapter	2	L-urbanophilic	H-urbanophilic	2
Passeridae					
<i>Passer domesticus</i> (House Sparrow)	Adapter	2	L-very urbanophilic	H-urbanophilic	2
<i>Passer montanus</i> (Tree Sparrow)	Adapter	2	L-urbanophilic	H-urbanophilic	2
<i>Petronia petronia</i> (Rock Sparrow)	Avoider	2			
<i>Montifringilla nivalis</i> (Snow Finch)	Avoider	2			
Fringillidae					
<i>Fringilla coelebs</i> (Chaffinch)	Adapter	1	L-moderately urbanophilic	H-moderately urbanophilic	1
<i>Serinus serinus</i> (Serin)	Adapter	1	L-urbanophilic	H-urbanophilic	1
<i>Serinus citrinella</i> (Citril Finch)	Avoider	2			
<i>Carduelis chloris</i> (Greenfinch)	Adapter	1	L-very urbanophilic	H-urbanophilic	1
<i>Carduelis carduelis</i> (Goldfinch)	Adapter	1	L-urbanophilic	H-urbanophilic	1
<i>Carduelis spinus</i> (Siskin)	Avoider	1			
<i>Carduelis cannabina</i> (Linnet)	Adapter	1	L-urbanophilic	H-urbanophilic	1
<i>Carduelis flammea</i> (Redpoll)	Avoider	1			
<i>Loxia curvirostra</i> (Crossbill)	Avoider	2			
<i>Pyrrhula pyrrhula</i> (Bullfinch)	Adapter	1			
<i>Coccothraustes coccothraustes</i> (Hawfinch)	Avoider	1			
Emberizidae					
<i>Emberiza citrinella</i> (Yellowhammer)	Avoider	1			
<i>Emberiza cirlus</i> (Cirl Bunting)	Avoider	1			
<i>Emberiza cia</i> (Rock Bunting)	Avoider	1			
<i>Emberiza hortulana</i> (Ortolan Bunting)	Avoider	1			
<i>Emberiza schoeniclus</i> (Reed Bunting)	Avoider	1			
<i>Miliaria calandra</i> (Corn Bunting)	Avoider	1			

32 cm wingspan). We classified traits into four categories: general biology, distribution, breeding, and morphometry.

Thirteen traits (10 qualitative and three quantitative) were initially selected; of these, main habitat, diet, social behavior, migratory vs. sedentary, nest openness, plumage dimorphism, nest height (m), maximum life expectancy (years), and number of molts per year for adults fell under the category of general biology. Faunistic category described species distribution; parental investment in nest building, parental investment in incubation, and date of the first clutch were considered species breeding parameters. Fifteen other traits (all quantitative) suspected to be correlated underwent Spearman rank correlation analysis. Based on those preliminary correlation results (Table 3), we retained five supplementary uncorrelated traits: number of pairs in France and global geographic range (km²) completed the description of

species distribution, mean number of clutches per year and the duration of time (days) that the young remain in nest after hatching completed the description of species breeding, and species wingspan was added as a descriptor of species morphometry. Finally, the each of 18 selected traits was declined into two to eight modalities (Table 2) to be included in the same cluster analysis. All modalities of traits were determined by taking both their biological significance and the number of species they described into account.

The numbers of species sampled at the regional and at the local scales were not similar. Consequently, we performed separate correlations for the regional and local species samples. For the same reason, some traits were declined in different modalities for the regional and the local analyses (e.g., nest height; Table 2). In addition, we excluded date of the first

TABLE 2. Description, by category, of traits selected for regional and local analyses. Trait modalities and the number of species per modality are indicated for each analysis scale. Eighteen traits were used for regional analysis and 17 for local analysis. The trait “date of the first clutch” was not considered in the local study because it showed insufficient variation among species surveyed along the rural-urban gradient (24 of 31 species have their first clutch in April).

Trait categories Traits	Nature of traits	Regional analysis		Local analysis	
		Modalities description	Number of species (<i>n</i> = 110)	Modalities description	Number of species (<i>n</i> = 31)
General biology					
Main habitat ^{a,b,c,d}	qualitative	shrubs	21	shrub	7
		forests	24	forests	12
		meadows	22	meadows	11
		rock faces (mountains, cliffs, other rock faces)	11	rock faces (mountains, cliffs, other rock faces)	1
		aquatic	12		
		open landscapes	20		
Diet ^{a,b,c,d}	qualitative	granivorous	23	granivorous	5
		insectivorous	71	insectivorous	20
		omnivorous	12	omnivorous	6
		carnivorous	4		
Social behavior ^d	qualitative	family	15	family	4
		groups	37	groups or small groups	22
		small groups	20	solitary	5
		solitary	35		
		uninformed	3		
Migratory vs. sedentary ^d	qualitative	sedentary	47	sedentary	24
		migratory (migratory or partially migratory)	63	migratory (migratory or partially migratory)	7
Nest height (m) ^{a,b,c,d}	quantitative	ground ($x \leq 0.5$)	32	ground or low ($x \leq 2$)	11
		low ($0.5 < x \leq 2$)	26	medium ($2 < x \leq 5$)	12
		medium ($2 < x \leq 5$)	23	high ($x > 5$)	8
		high ($x > 5$)	26		
		variable (from the ground to high)	3		
Nest openness ^d	qualitative	open	70	open	15
		enclosed (ovoid nest or cavity)	40	enclosed (ovoid nest or cavity)	16
Maximum life expectancy ^d	quantitative	3 years	27	3 years	14
		5 years	71	5 years and more	17
		10 years and more	8		
		uninformed	4		
Number of molts per year for adults ^b	quantitative	1 molt	65	1 molt	27
		2 molts	45	2 molts	4
Plumage dimorphism ^c	qualitative	yes	46	yes	11
		no	64	no	20
Distribution					
Number of pairs in France ^f	quantitative	least abundant ($x < 10\,000$)	17	not very abundant ($x < 1\,000\,000$)	13
		not very abundant ($10\,000 \leq x < 100\,000$)	30	moderately abundant ($1\,000\,000 \leq x < 4\,000\,000$)	11
		moderately abundant ($100\,000 \leq x < 1\,000\,000$)	26	very abundant ($x \geq 4\,000\,000$)	7
		abundant ($1\,000\,000 \leq x < 2\,000\,000$)	20		
		very abundant ($x \geq 2\,000\,000$)	15		
		uninformed	2		

(continued)

TABLE 2. (Continued)

Trait categories Traits	Nature of traits	Regional analysis		Local analysis	
		Modalities description	Number of species (<i>n</i> = 110)	Modalities description	Number of species (<i>n</i> = 31)
Faunistic category ^e	qualitative	European	19	European	8
		Palearctic	37	Palearctic	14
		European-Turkestanian	17	Others	9
		Holarctic	9		
		Mediterranean	10		
Global geographic range (km ²) ^e	quantitative	Others: Old World, Indian-African, Palaeomontane, Palaeoxeric, Palaeo-xeromontane, Siberian, Turkestanian-Mediterranean	18		
		narrow ($x < 10\,000\,000$)	38	moderate ($x < 15\,000\,000$)	13
		moderate ($10\,000\,000 \leq x < 20\,000\,000$)	41	wide ($15\,000\,000 \leq x < 20\,000\,000$)	7
		wide ($x \geq 20\,000\,000$)	31	very wide ($x \geq 20\,000\,000$)	11
Breeding Parental investment in nest building ^{a,b,c,d}	qualitative	female alone	62	female alone	17
		both parents	46	both parents	14
Parental investment in incubation ^{a,b,c,d}	qualitative	uninformed	2		
		female alone	89	female alone	26
Date of the first clutch ^{a,b,c,d}	qualitative	both parents	21	both parents	5
		March	10		
Mean number of clutches per year ^{a,b,c,d}	quantitative	April	59		
		May	41		
		up to 2 clutches ($x \leq 2$)	61	fewer than 2 clutches ($x < 2$)	16
Duration young remain in nest after hatching (days) ^{a,b}	quantitative	more than 2 clutches ($x > 2$)	49	2 clutches or more ($x \geq 2$)	15
		short duration ($x \leq 14$)	49	short duration ($x \leq 15$)	14
		mean duration ($14 < x \leq 16$)	22	long duration ($x > 15$)	17
		long duration ($x > 16$)	36		
Morphometry Wingspan (cm) ^{b,d}	quantitative	uninformed	3		
		small ($x \leq 20$)	32	small ($x \leq 20$)	14
		medium ($20 < x \leq 25$)	31	large ($20 < x \leq 30$)	9
		large ($25 < x \leq 30$)	22	very large ($x > 30$)	8
		very large ($x > 30$)	25		

^aYeatman-Berthelot and Jarry (1994).^bGéroutet (1957-1972, 1961, 1963).^cMullarney et al. (1999).^dHume et al. (2003).^eVoous (1960).^fDubois et al. (2001).

clutch from the traits used in the local scale analysis, as it showed little variation among species surveyed (three species in March, 24 species in April, and four species in May). In total, 18 traits were included in the regional analysis and 17 in the local analysis (Table 2).

STATISTICAL ANALYSES

Using program SPAD (Lebart et al. 2000), we performed cluster analyses (multiple correspondence analysis followed by a hierarchical technique of classification) on the biological traits of species sampled in regional and local studies. Cluster

analyses were based on Euclidean distances. In both local and regional analyses, species status was used as an illustrative variable. Consequently, groups of species defined by cluster analysis were based only on similarities of species according to their biological traits. Results give groups of species and modalities of traits that significantly characterize each group (i.e., modalities significantly more frequent among species belonging to a group than among the total species pool). Significance is based on the test value and its associated P -value calculated by program SPAD (95% CI; Lebart et al. 2000). Species tolerance was used only to illustrate if one of these species groups was significantly constituted by species of one particular tolerance status. If so, we concluded that species of that status were more prone to present trait modalities that characterized the considered group and that those modalities could help to explain that species tolerance level to urban areas.

RESULTS

TRAITS OF URBAN ADAPTERS VS URBAN AVOIDERS

Study at the regional scale involved 110 passerine species belonging to 22 bird families. Sixty-five species (59%) were urban avoiders, and 45 (41%) were urban adapters (Table 1). No exotic species were observed in these surveys. Cluster analysis identified a minimum of two groups of species. The first group was significantly associated with urban avoider status ($P = 0.001$; Table 4). It contained 72 species, 51 (71%) of which were avoiders, and 21 (29%) of which were adapters. The second group was significantly associated with urban adapter status ($P = 0.001$; Table 4). It contained 38 species, 14 (37%) of which were avoiders, and 24 (63%) of which were adapters.

Urban adapters were significantly associated with 16 traits and 19 modalities. Urban avoiders were significantly described by 15 traits and 20 modalities, all opposite to those that described urban adapters (Table 4). We therefore describe only urban adapters here. The main habitat of urban adapter birds corresponds to mountains or cliffs and to forest. These birds are mainly omnivorous. They live in groups larger than their own families and are sedentary species. They build their nests up to 2 m high, and nests are generally enclosed. These bird species have high life expectancies. Adults molt once a year and have no plumage dimorphism. Urban adapter species are principally Holarctic and widely globally distributed species. Concerning their reproduction, both parents build nests, first clutches are in April, species have a mean of two clutches maximum per year, and young remain in nests for a long time (≥ 16 days) after hatching. With correlation results (Table 3), we can also consider these urban adapters to have larger numbers of eggs per clutch, longer incubation times, and

shorter nesting times than urban avoiders. Urban adapters are also species with large wingspans, suggesting, according to correlation results, that they are bigger than urban avoider species.

Only two traits—number of pairs in France and parental investment in incubation—had no modalities significantly associated with urban adapters or avoiders. Global geographic range had no modality associated with urban avoiders, suggesting that these species are equally widely or narrowly distributed.

TRAITS OF INDICATOR SPECIES ALONG THE RURAL-URBAN GRADIENT

Thirty-one species were recorded along the rural-urban gradient. All of them were considered urban adapters in the regional study (Table 1). According to the PCA on vegetation within woodlands, the first three principal components, PC1, PC2, and PC3, accounted for 43%, 19%, and 15% of the total inertia, respectively (Fig. 2A). Both PCA and HAC grouped sites according to similarities in vegetation structure but did not highlight a precise rural-urban dichotomy. The IndVal analysis identified the most numerous indicator species when considering cluster partition with four groups (Fig. 2B and 2C). The first group corresponded to the large urban site 00, characterized by a large shrubby layer, poor litter coverage, and a poor and undiversified herbaceous layer. No indicator species were identified for this site. The second group corresponded to the urban park 01. This site had similar characteristics to site 00 but had a richer herbaceous layer. Swallows (*Hirundo rustica*) and Firecrest (*Rugulus ignicapillus*) were identified as indicators of this urban site and were considered H-very urbanophilic species. The IndVal analysis identified no indicator species for the third group of the cluster analysis involving sites 03, 04, 05, 06, and 12. The fourth group of the cluster analysis involved one urban site (02) and five rural sites (07, 08, 09, 10, and 11), all characterized by an important tree coverage and bare ground underneath. The European Robin (*Erithacus rubecula*), Chiffchaff (*Phylloscopus collybita*), Blue Tit (*Parus caeruleus*), and Chaffinch (*Fringilla coelebs*) were identified as indicator species of this fourth group and were considered H-moderately urbanophilic species. All of the 25 other species were simply considered H-urbanophilic.

According to the landscape PCA (Fig. 3A), PC1 and PC2 accounted for 67% and 17% of the total inertia, respectively. Rural sites appear on the right of PC1, and urban sites are on the left. Both PCA and HAC distinguished urban sites from rural ones. Urban landscapes show higher fragmentation, more diverse land use, more mineral surfaces, smaller and more irregular patches, less crop cover, and less connectivity between woods than rural sites (Fig. 3B). IndVal analysis identified six species as indicators of urban sites (Fig. 3C): the Eurasian

TABLE 3. List of 15 quantitative traits describing distribution, breeding, or morphometry of species and results of the Spearman rank correlations carried out between those traits within categories. Asterisks designate traits retained for further analyses of clustering according to these results; all others traits were excluded.

Trait categories Tested traits	Regional analysis		Local analysis	
	r_s	P	r_s	P
Distribution				
*Global geographic range (km ²)—global latitudinal range from north to south (km)	0.83	<0.001	0.78	<0.001
*Global geographic range (km ²)—*number of pairs in France	0.14	0.44	0.14	0.44
Breeding				
*Mean number of clutches per year—*time the young stay in nest after hatching (days)	−0.34	<0.001	−0.47	0.01
*Mean number of clutches per year—mean number of eggs per clutch	−0.28	0.004	−0.59	0.001
*Mean number of clutches per year—nesting time (months)	0.53	<0.001	0.66	<0.001
*Mean number of clutches per year—incubation time (days)	−0.39	<0.001	−0.43	0.02
*Time the young stay in nest after hatching (days)—mean number of eggs per clutch	0.26	0.008	0.55	0.003
*Time the young stay in nest after hatching (days)—nesting time (months)	0.07	0.48	−0.17	0.37
*Time the young stay in nest after hatching (days)—incubation time (days)	0.66	<0.001	0.59	0.001
Mean number of eggs per clutch—nesting time (months)	−0.15	0.14	−0.48	0.01
Mean number of eggs per clutch—incubation time (days)	0.21	0.03	0.37	0.05
Nesting time (months)—incubation time (days)	<0.01	0.93	<−0.01	1.00
Morphometry				
*Wingspan (cm)—folded wing length (mm)	0.95	<0.001	0.91	<0.001
*Wingspan (cm)—tail length (mm)	0.63	<0.001	0.75	<0.001
*Wingspan (cm)—beak length (mm)	0.62	<0.001	0.70	<0.001
*Wingspan (cm)—tarsus length (mm)	0.53	<0.001	0.48	0.01
Folded wing length (mm)—tail length (mm)	0.61	<0.001	0.65	<0.001
Folded wing length (mm)—beak length (mm)	0.64	<0.001	0.72	<0.001
Folded wing length (mm)—tarsus length (mm)	0.53	<0.001	0.48	0.008
Tail length (mm)—beak length (mm)	0.54	<0.001	0.59	0.001
Tail length (mm)—tarsus length (mm)	0.67	<0.001	0.71	<0.001
Beak length (mm)—tarsus length (mm)	0.79	<0.001	0.66	<0.001
*Wingspan (cm)—length (cm)	0.87	<0.001	0.91	<0.001
*Wingspan (cm)—weight (g)	0.90	<0.001	0.90	<0.001
Length (cm)—weight (g)	0.92	<0.001	0.93	<0.001

Blackbird (*Turdus merula*), Song Thrush (*Turdus philomelos*), Goldcrest (*Regulus regulus*), Magpie (*Pica pica*), House Sparrow (*Passer domesticus*), and Greenfinch (*Carduelis chloris*). These species were considered L-very urbanophilic. Four other species were identified as indicators of rural sites and were considered L-moderately urbanophilic: the Chiffchaff, Blue Tit, Great Tit (*Parus major*), and Chaffinch. All of the 22 other species were simply considered L-urbanophilic.

Based on biological traits, cluster analysis identified a minimum of two groups of species (Table 5). Groups 1 and 2

contained 13 and 18 species, respectively. Species of the first group were characterized by six traits and six modalities. These species are granivorous, with individuals staying together in families and having open nests. These species are moderately or widely distributed. Females build nests alone, and the duration of time that the young remain in nests after hatching is short (≤ 15 days). Species of the second group were characterized by nine traits and nine modalities. Individuals stay in groups larger than just their families, their nests are enclosed, and they are more widely distributed than species

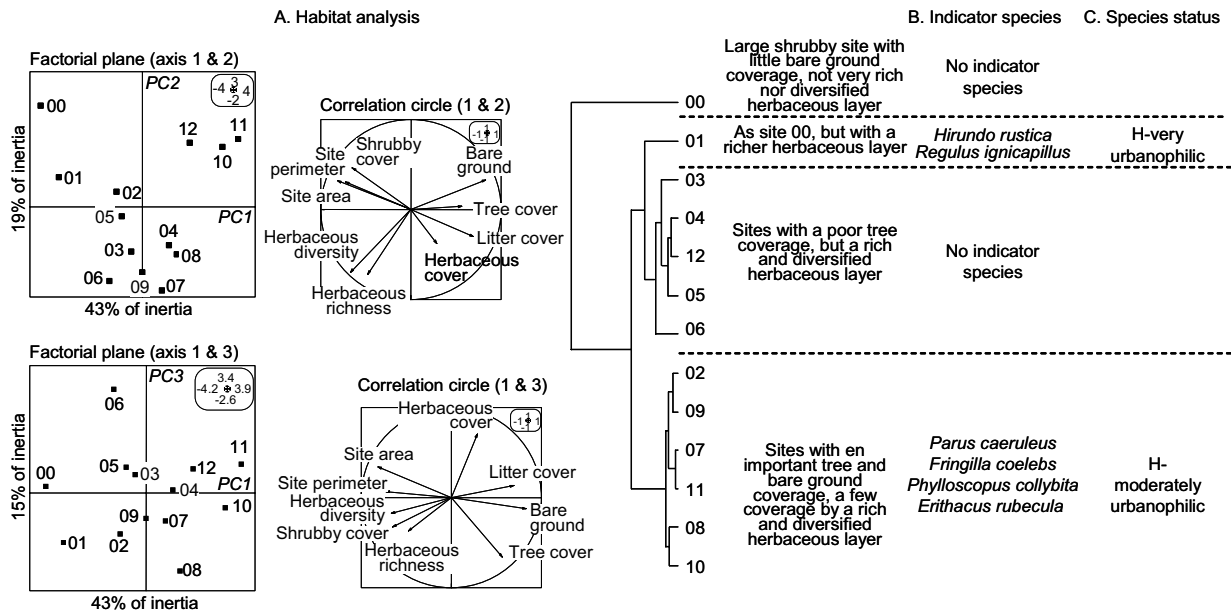


FIGURE 2. Results of the IndVal analysis carried out on data on vegetation structure within woodlands. (A) Habitat analysis: PCA (factorial planes PC1-PC2, and PC1-PC3) and hierarchical ascendant classification highlighted different groups of sites in terms of their vegetation. (B) IndVal analysis: among 31 species, two were identified as indicators of the urban site 01, and four were identified as indicators of sites 02, 07, 08, 09, 10, and 11. (C) Species status: indicator species of urban site 01 were associated with H-very urbanophilic status because they appeared very tolerant to the vegetation structure within an urban woodland. Indicator species of sites 02, 07, 08, 09, 10, and 11 were associated with H-moderately urbanophilic status because they appeared slightly tolerant to the vegetation structure within urban woodland (only site 02 is urban). All other species ($n = 25$) were associated with H-urbanophilic status because they tolerated vegetation structure within both urban and rural woodlands.

of the first group. The young remain longer in the nest after hatching, both parents build nests, and breeding individuals have fewer than two clutches per year. However, none of these groups was significantly associated with very urbanophilic, urbanophilic or moderately urbanophilic status based either on habitat or landscape descriptions. Consequently, we could not associate biological traits with any of the indicator species identified along our rural-urban gradient.

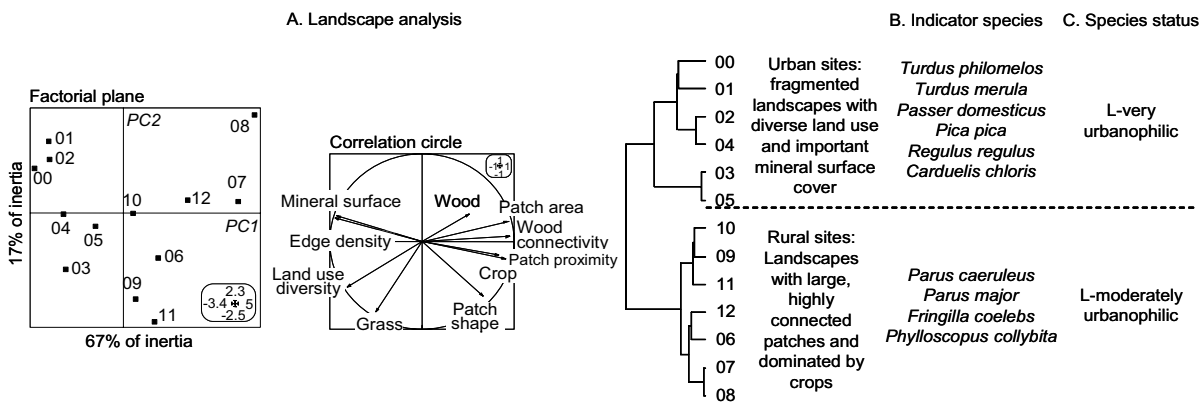


FIGURE 3. Results of the IndVal analysis carried out on data on landscape composition and configuration in a buffer zone with a radius of 600 m around each woodland. (A) Landscape analysis: PCA (factorial plane PC1-PC2) and hierarchical ascendant classification highlighted differences between landscapes inside and outside the city. (B) IndVal analysis: among 31 species, six were identified as indicators of urban sites 00, 01, 02, 03, 04, and 05; and four were identified as indicators of rural sites 06, 07, 08, 09, 10, 11, and 12. (C) Species status: indicator species of urban sites were associated with L-very urbanophilic status because they appeared very tolerant to urban landscape. Indicator species of the rural sites were associated with L-moderately urbanophilic status because they appeared slightly tolerant to urban landscapes. All others species ($n = 21$) were associated with L-urbanophilic status because they tolerate both rural and urban landscapes.

TABLE 5. Results of the cluster analysis of the local study. Cluster analysis discriminated two groups of species on the basis of their biological traits (active traits). Habitat tolerance status, and landscape tolerance status (illustrative traits) were used to illustrate the amount of their associated modalities in each species group. Illustrative and active modalities describing species groups are given with their associated frequencies and *P*-values. "Modality frequency (%) in the group" gives the percent of species associated with a modality among species involved in the first ($n = 13$) or second ($n = 18$) species group. "Modality frequency (%) in the total species sample" gives the percent of species associated to a modality among all species of the total sample (i.e., involved in the local analysis; $n = 31$). *P*-values indicate whether a modality is significantly more frequent among species belonging to a group or than among the total species sample.

		First local species groups ($n = 13$ species)				Second local species groups ($n = 18$ species)			
	Trait categories Traits	Modalities	Modality frequency (%) in group	Modality frequency (%) in total species sample	<i>P</i>	Modalities	Modality frequency (%) in group	Modality frequency (%) in total species sample	<i>P</i>
Illustrative	Habitat tolerance status	H-moderately urbanophilic	23	13	0.19	H-very urbanophilic	11	6	0.33
	Landscape tolerance status	L-moderately urbanophilic	15	13	0.43	H-urbanophilic	83	81	0.50
		L-very urbanophilic	23	19	0.50	L-urbanophilic	72	68	0.40
Active	General biology								
	Main habitat	shrub	38	23	0.09	forest	56	39	0.03
		meadows	46	35	0.25	rock faces	6	3	0.42
	Diet	granivorous	38	16	0.008	insectivorous	67	65	0.47
						omnivorous	33	19	0.03
	Social behavior	family	31	13	0.02	groups or little groups	89	71	0.01
	Migratory or sedentary	solitary	23	16	0.34				
		migratory	31	23	0.31	sedentary	83	77	0.31
	Nest height (m)	ground or low	54	35	0.08	medium	44	39	0.35
						high	33	26	0.24
	Nest openness	open	77	48	0.009	enclosed	72	52	0.009
	Maximum life expectancy (years)	5 years	69	55	0.16	3 years	56	45	0.16
		and more							
	Number of molts per year for adults	2 molts	23	13	0.19	1 molt	94	87	0.19
	Plumage dimorphism	yes	46	35	0.25	no	72	65	0.25
	Distribution								
	Number of pairs in France	moderately abundant	46	35	0.25	not very abundant	56	42	0.07
		very abundant	31	23	0.31				
	Faunistic category	European	38	26	0.17	others	61	45	0.04
		Palaearctic	38	29	0.28				
	Global geographic range (km ²)	moderate	54	42	0.22	very wide	61	35	<0.001
		wide	46	23	0.01				
Breeding	Parental investment in nest building	female alone	92	55	<0.001	both parents	72	45	<0.001
	Parental investment in incubation	female alone	92	84	0.29	both parents	22	16	0.29
	Mean number of clutches per year	2 clutches	69	48	0.05	fewer than 2 clutches	67	52	0.05
		or more							
	Duration young remain in nest after hatching (days)	short duration	100	45	<0.001	long duration	94	55	<0.001
Morphometry									
	Wingspan (cm)	large	46	29	0.08	small	56	45	0.16
						very large	28	26	0.45

DISCUSSION

URBANIZATION AS AN ENVIRONMENTAL FILTER: SCALE DEPENDENCE

Our results support previous studies in highlighting that life history strategies (i.e., here, combinations of biological traits) differ between urban adapters and urban avoiders, especially concerning habitat, diet, sociability, and migratory status (Lancaster and Rees 1979, McClure 1989, Lim and Sodhi 2004, Clergeau, Croci, et al. 2006, Devictor et al. 2007, Kark et al. 2007). Our results also highlight other original differences between urban adapters and urban avoiders such as the number of molts per year and plumage dimorphism. In addition, our results show that urbanization acts as a biological filter at a regional but not local scale. Indeed, several biological traits differed between urban adapters and urban avoiders (regional analysis); however, among urban adapters, no traits differed between very urbanophilic and moderately urbanophilic species (local analysis). Differences in several factors between these analyses can explain the different results, such as the number of habitats involved for species samples and the spatial scale considered.

As our results show, no single trait can discriminate between urban avoiders and urban adapters. We thus agree with Kark et al. (2007) that only the combination of several trait modalities can reliably indicate if a species is a potential urban avoider or adapter. Several traits identified in our study can help in such discrimination; however, our results do not enable us to determine how many traits are essential to differentiate between urban avoiders and urban adapters. Doing so would require identification and quantification of all possible correlations and trade-offs among traits.

In the regional analysis, we took into account species encountered in a large landscape matrix with heterogeneous and diverse land uses. In the local analysis, we focused on woodlands located in an increasingly urbanized matrix. Among the 45 urban adapters identified in the regional analysis, only two species preferred open landscapes or aquatic habitats. Urban adapters were mainly species that preferred forest (15 of 45 species) or meadow (14 of 45 species) habitats. This preference could explain why all species sampled in woodlands along the gradient were actually urban adapters—they all can potentially find suitable habitat in cities. This observation suggests that habitat availability in cities is a key determinant of the biological filter that urban constraints play on bird species from natural areas. It also illustrates why results of studies on bird communities along a rural-urban gradient can depend on whether several habitats (as in our regional analysis) or a single habitat type (as in our local analysis) have been considered (Jokimäki and Kuisanlahti-Jokimäki 2003, Clergeau, Jokimäki, and Snep 2006, Sadler et al. 2006).

However, none of the 23 urban avoiders exploiting forest, rock faces, or meadow habitat (that are available in towns) were detected in our woodlands. Thus, it appears that species detected in woodlands located in a rural context—at 10 m from

the town center—have already been subject to biological filtration. Therefore, in addition to habitat availability, other factors may determine the biological filtering effect of urban constraints. First, woodland areas range from 0.5 to 2 ha, which may not be sufficiently large for some species. As Natuhara and Imai (1996) suggested, some species need deep forest and do not inhabit small fragments of forests. Second, urbanization is more complex in some habitat and landscape variables (Husté et al. 2006) than can be fully described by our rural-urban gradient. Some urban constraints we have not yet identified, such as pollution, might affect species beyond urban area frontiers (i.e., at a wider scale than a town area).

As shown by the IndVal analysis, most of our species in woodlands were equally distributed along the rural-urban gradient, and differences in distributions of the few indicator species were more related to landscape and habitat features than to their life history strategies. Thus, composition and structure of bird communities varied slightly when focusing only on woodlands along the gradient. In addition, these results suggest that the biological filter that may be operating between towns and more natural landscapes at a wide scale may not be the one causing the specific distribution of birds along a short rural-urban gradient when focusing on a single habitat type. Along such a gradient, landscape modifications or variations in habitat quality seem to be more relevant factors explaining species distribution variations (Clergeau et al. 2001, Mörtberg 2001, Melles et al. 2003, Donnelly and Marzluff 2006, Sandström et al. 2006).

Urban areas seem to impose so many constraints on birds (e.g., landscape fragmentation, isolation of habitat patches, noise, pollution, type and availability of food resources, human activities, and vegetation quality in habitat patches) that only species that have (pre-) adapted biological traits are able to tolerate them (Blair 1996, Kark et al. 2007). In that sense, we can really consider urbanization to filter bird species on the basis of their traits (Fig. 4). Urban constraints are common in towns (Clergeau et al. 2001); the same life history strategies (i.e., the same traits) tend to be selected in towns, thus leading to a functional homogenization of passerine communities by urbanization (Clergeau, Croci, et al. 2006, McKinney 2006). The biological filtering effect of urbanization seems to be a key determinant in which passerines can be found in town (urban adapters) and which cannot (urban avoiders). Biological traits associated with urban adapters may offer advantages to birds in urban environments (descriptions below).

ADVANTAGES OF TRAITS OF URBAN ADAPTERS VS. URBAN AVOIDERS

The abundance of species is not the main factor explaining their presence in cities. Indeed, urban adapters and avoiders were not associated with any of the modalities of the number of species pairs in France trait. Geographic distributions of urban adapters were especially wide and involved more diverse

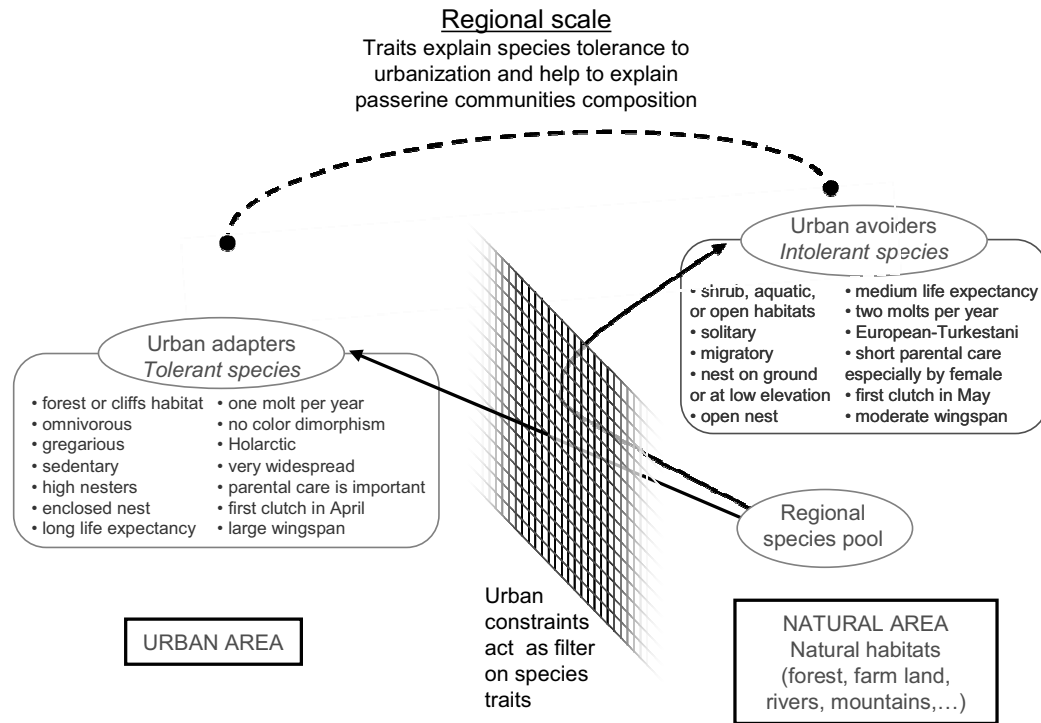


FIGURE 4. Within the regional species pool, some species appear tolerant to urban areas, and others appear intolerant. These opposite tolerance patterns are linked to different biological traits that could enhance species tolerance to urban constraints. At a regional scale, urbanization seems to act as a filter on species traits. These traits could help to identify a priori urban avoiders from urban adapters and to understand differences in composition between urban and rural communities on a large spatial scale.

biogeographic areas (principally Holarctic but also Siberian and Indian-African species) than urban avoiders, which were generally from temperate climates (European, European-Turkestanian, and Mediterranean). Being a widespread species (i.e., a species that has already encountered a variety of environmental conditions) appears to favor tolerance to urban constraints (Crocini et al. 2007).

In cities, we found a decrease in species preferring shrub habitats (Lancaster and Rees 1979, Lim and Sodhi 2004, Clergeau, Crocini, et al. 2006) and aquatic or open landscapes, and an increase of species preferring forest habitats (McClure 1989) and cliffs and mountains. These latter species can be favored in town because aquatic or open landscapes are rare there, while urban forests are more common and similar to natural forests. In addition, urban buildings may compensate for the lack of the natural habitat of cliff-dwelling species such as swallows (Lim and Sodhi 2004).

Ground-nesting species seem to be at a disadvantage in French cities. Most species with high nests are associated with tree availability and therefore forest habitats, whereas nesting at low or medium heights is often correlated with shrubby (Leston and Rodewald 2006), aquatic, or open landscapes. Habitat availability could act as a covariate of nest height, but high nesting can also constitute a real advantage for species. Martin and

Badyaev (1996) showed that predation rates on bird nests are lowest in the canopy. Therefore, high nest placement should make it possible to avoid predation pressure or human disturbances in cities (Fernández-Juricic and Tellería 2000, Jokimäki and Huhta 2000). Our enclosed-nest trait included cavity nests, and like Lancaster and Rees (1979), we found that cavity nesters are at an advantage in cities. Such nests may also provide some protection for clutches. These results, however, have not been supported by other studies, which found that cavity nesters were at a disadvantage in urban areas (Lim and Sodhi 2004, Blewett and Marzluff 2005).

Similarly to Lancaster and Rees (1979), Clergeau et al. (1998), and Jokimäki and Suhonen (1998), we found that omnivorous species are at an advantage in French cities, whereas granivorous species are at a disadvantage. Omnivorous species can eat a variety of food items, which might be an advantage in cities, where human activities mass produce novel resources such as garbage (Clergeau et al. 1998).

Unlike urban avoiders, urban adapters do not especially present dimorphism in plumage coloration. Sexually dimorphic plumage is expected to be more common among insectivorous or granivorous species with carotenoid-poor diets, for which acquiring carotenoids could be considered an honest signal of male quality (Gray 1996). Contrary to Kark et al. (2007), we found

that granivory was especially associated with urban avoiders and, therefore, plumage dimorphism. According to McLain et al. (1999), dimorphism in bird species could indicate that an important part of their energy is allocated to sexual selection, to the detriment of other adaptations such as survival in new environments. Although the species we surveyed in our regional approach are not very colorful, this hypothesis could provide an explanation for the tolerance or capacity to adapt to the new environment that cities represent. Moreover, plumage dimorphism is generally associated with reduced care of offspring by males. McLain et al. (1999) noted that reduced male care seemed to adversely affect “clutch size, . . . limit body size . . . **and therefore, competitive ability . . . and increased nest predation rates.**” Urban avoiders generally have more than two clutches per year (with few eggs per clutch, because number of clutches and clutch size is negatively correlated) and smaller wingspans compared to urban adapters. These results support the hypothesis that, compared to urban adapters, urban avoiders have reduced ability to adapt, compete, and reproduce in cities.

Urban adapters seem to have longer life expectancies than urban avoiders, which is difficult to explain. It could be interesting to study whether longevity compensates for low reproduction rates in urban birds, which seem to invest more in survival than in reproduction, as suggested by their lack of dimorphism.

Urban adapters also appear to be gregarious species (McClure 1989, Kark et al. 2007). Being with conspecifics may enhance opportunities to learn from others birds' experiences, such as in food location or predation avoidance, and thus be an advantage in new environments such urbanized areas.

As in others studies, we found that migrant species seem to be at a disadvantage in urban environments, and sedentary species at an advantage (McClure 1989, Dunford and Freemark 2004, Lim and Sodhi 2004, Kark et al. 2007). Sedentary species might already be adapted to climatic conditions of the region where a city is located, and they have more time and opportunities to be confronted with urban constraints and to adapt to them than do migrant species. In addition, urban avoiders generally have their first clutches in May (i.e., later than urban adapters), which may be due to the late arrival in the season of some migrant species. Sedentary birds, which spend their entire lives in the same environments, seem to be better adapted to their environments, where they are better competitors for resources than migrant species (Jokimäki and Suhonen 1998).

Urban adapters molt once a year, whereas urban avoiders molt twice yearly. Loss of old feathers creates gaps in wings and thus reduces wing area, resulting in changes in escape flight efficiency (Lind 2001) and flying performance (Swaddle and Witter 1997). Like reproduction, this renewal of feathers requires energy and nutrients; consequently, researchers have suggested a trade-off between molting and reproduction, especially in migratory species (Hemborg et al. 2001). Adults that molt while raising their young reduce their parental efforts such as feeding their young (Moreno et al. 2001), and increase their predation risk,

resulting in fitness costs (Hemborg and Lundberg 1998). Molting especially overlaps with breeding in migrant species, leaving a smaller time window for breeding before migration (Jenni and Winkler 1994). Migrant species might thus suffer more from predation and fitness costs than do sedentary species. Lind (2001) suggested that because they are not temporally constrained by migration, sedentary may molt more slowly than migrant species. A slow molt may largely not affect flight ability, and the reduced predation risk (compared to fast-molting species) that results may be a benefit of being sedentary. With regard to this trade-off between molt and reproduction in migrant species, we can better understand why urban adapters are especially sedentary species with one molt per year, and can have greater parental investment in nest building, longer fledgling care after hatching (more than 16 days), and more eggs per clutch (negatively correlated with number of clutches per year) than urban avoiders. All these traits lead us to believe that during the breeding season, urban avoiders have fewer resources than urban adapters to adapt to new environments such as urban areas.

In summary, with regards to other studies on urban birds, urban adapters are species that seem to allocate higher energy to survival and for which reproduction has a lower relative cost than for urban avoiders. Urban adapters seem to prefer forested habitat, are relatively sedentary, omnivorous, widely distributed, high nesters, and have large wingspans (Fig. 4).

MANAGEMENT IMPLICATIONS

Urbanization is spreading throughout the world, always at the expense of natural areas (Germaine and Wakeling 2001, McKinney 2006). Consequently, it is a very important challenge for biological conservation and urban planning to identify species that will be preadapted to urban constraints and might become pest species, or those that will not be able to survive in an urban setting. In accordance with other studies, our results at a regional scale highlight that measurable and accessible traits can allow such identification. This could help, for example, with decisions about orientation of urban areas to preserve a species located near a town and suspected to be an urban avoider, or in the management of an exotic species that recently arrived in the surroundings of a town that is suspected to be an urban adapter. Our study on traits also highlights the importance of the availability of diverse habitat types. Town planners should try to integrate or preserve diverse natural habitats such as large forests or shrubby, open, and aquatic landscapes into future urban extensions (Natuhara and Imai 1996, Fernández-Juricic 2004, Kark et al. 2007). For example, brownfields should be maintained in cities, and riversides should also be kept in as natural condition as possible instead of being highly managed. Finally, urban managers should also act on landscapes around woodlands and on habitat quality within woodlands to make urban passerine communities (i.e., the occurrence and abundance of species) more like rural passerine ones. For example, developing shrubs and avoiding

bare ground in urban woodlands, and increasing woodland connectivity in urban areas may help species we identified as H- or L-moderately urbanophilic to reach the same densities in urban woodlands as in rural woodlands.

Our results support the hypothesis that urbanization filters birds on the basis of their traits at wide spatial scales. Only species that have (pre-) adapted life history strategies can tolerate urban constraints and thus exploit urban areas. Urban constraints that may filter species seem to involve habitat availability in town and can act beyond urban areas, and do not explain local distribution of species along a rural-urban gradient. These results encourage analyses of biological trait to enhance our understanding of ecological processes as the basis of community organization (McGill et al. 2006).

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